

NADEN MOHAMED YASEN

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EDUCATION

Helwan University

B.Sc. Computer Science & Artificial Intelligence · GPA: 3.37/4.0

Cairo, Egypt

Sep 2022 – Jun 2026

SKILLS

Languages: Python, C, C++, Java

ML & AI: Scikit-learn, PyTorch, TensorFlow, Natural Language Processing (NLP), Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), GenAI, Fine-Tuning, Prompt Engineering

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Dash

Frameworks & Tools: FastAPI, Hugging Face, LangChain, LangGraph, Git, GitHub, Jupyter, Google Colab, VS Code, Postman, Docker, Streamlit, Gradio, Beautiful Soup, Selenium

Databases & Systems: FAISS (Vector Search), SQL, MongoDB Atlas, Qdrant, Postgres, Pgvector, SQLAlchemy, DBaver

Core Concepts: Statistics, Linear Algebra, Data Structures, Algorithms, OOP, System Design, Clean Architecture, Exploratory Data Analysis (EDA), Model Evaluation

Languages Spoken: Arabic, English

PROJECTS

Toxic Content Classification System | PyTorch, DistilBERT, LoRA, Llama Guard, BLIP

- Built and evaluated an LSTM-based toxic text classifier using PyTorch, achieving measurable accuracy, precision, recall, and F1-scores across multi-label toxicity categories.
- Fine-tuned DistilBERT using Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA), reducing training cost and memory usage while maintaining classification performance tracked via Weights & Biases (W&B).
- Integrated a zero-shot content moderation pipeline using Llama Guard with structured prompt engineering, alongside a multimodal BLIP image captioning module, deployed end-to-end via Streamlit with MongoDB Atlas for logging.

Healthcare RAG Chatbot | Python, LangChain, LangGraph, FAISS, Streamlit

- Developed a Healthcare RAG Chatbot leveraging the Groq API, enabling accurate, context-aware medical Q&A.
- Implemented multi-source retrieval from Wikipedia API, DuckDuckGo, and multi-format document uploads to enhance chatbot knowledge base.
- Engineered a content preprocessing pipeline with HuggingFace embeddings and FAISS vector store for efficient semantic search.
- Defined a hybrid retrieval scoring mechanism combining semantic similarity with credibility weighting, improving retrieval accuracy by 10%.
- Designed a Streamlit interface with LangGraph enabling structured multi-turn conversations and improved user interaction flow.

Egypt Tourism Sentiment Analysis | TensorFlow, XGBoost, LangChain, Dash

- Created a sentiment analysis pipeline to analyze foreign tourist opinions on Egypt using scraped 30k+ tourism reviews from TripAdvisor and Quora.
- Conducted text preprocessing, handled class imbalance, and implemented ML models using TensorFlow: Logistic Regression (89.6% accuracy), achieved 90.7% accuracy using XGBoost, outperforming baseline models.
- Integrated Google Gemini Flash API with LangChain to generate actionable recommendations.
- Visualized insights using word clouds, confusion matrices, and a Dash dashboard; deployed models with Streamlit.

TRAINING & CERTIFICATIONS

IBM-Certified AI & Data Science Professional | Digital Egypt Pioneers Initiative (DEPI) Nov 2024 – May 2025

- Completed an IBM-certified AI & Data Science program covering Python, ML, SQL, and MLOps.
- Built and evaluated machine learning and deep learning models, achieving 91.12% accuracy on the capstone project.

NLP Training | Neurova AI

Apr 2026 – Jul 2026

- Completed an intensive 8-week NLP training program covering the modern NLP stack from classical preprocessing and TF-IDF through Transformer architectures, BERT family fine-tuning, and LLM integration.
- Delivered 7 end-to-end NLP projects — applying skills in text classification, semantic search, vector databases, FastAPI integration, PEFT fine-tuning using LoRA, LangChain, and LangGraph.